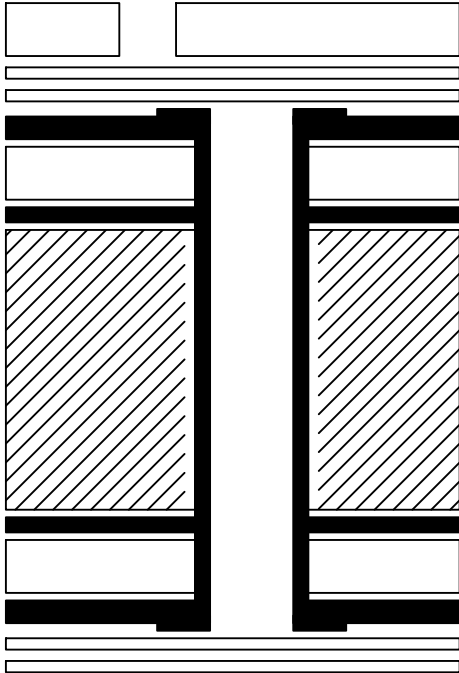


Board Specifications

Layer Stackup



Layer Name	Type	Material	Thickness
F.Paste	Solder Paste		0.10 mm
F.Silkscreen	Silk Screen	Not specified	Not specified
F.Mask	Solder Mask	Not specified	Not specified
L1(SIG)	Copper		0.035 mm
Dielectric 1	Prepreg	3313	0.0994 mm
L2 (GND)	Copper		0.0152 mm
Dielectric 2	Core	FR4 TG155	1.265 mm
L3 (GND)	Copper		0.0152 mm
Dielectric 3	Prepreg	3313	0.0994 mm
L4(SIG)	Copper		0.035 mm
B.Mask	Solder Mask	Not specified	Not specified
B.Silkscreen	Silk Screen	Not specified	Not specified

Specify Stackup: JLC04161H-3313

DESIGN GEOMETRY MINIMUM FEATURE SIZES

Min. Track Clearance	0.15 mm
Min. Track Width	0.15 mm
Min. Drill Hole	0.30 mm
Min. Annular Ring	0.10 mm
Min. VIA Diameter	0.50 mm
Copper to Hole Clearance	0.25 mm
Copper to Edge Clearance	0.30 mm
Hole to Hole Clearance	0.25 mm
Min. Text Height	1.00 mm
Min. Text Thickness	1.50 mm

Project: 2-Phase Interleaved Sync-Buck Converter		LQRx	
Title: Fabrication Documents	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:
	Reviewed by:	Date:	Rev: 1.1 Date: 28/06/2026
	Sheet: Stackup	File: 2Phase_SynchBuck.kicad_pcb	Size A4 Sheet: 1 of 5
KiCad E.D.A. 10.0.6			

Top Assembly (Scale 2:1)

28V MAX

20A MAX

x2

USB

SWD

AUX

RST

FLT

ON

DAC

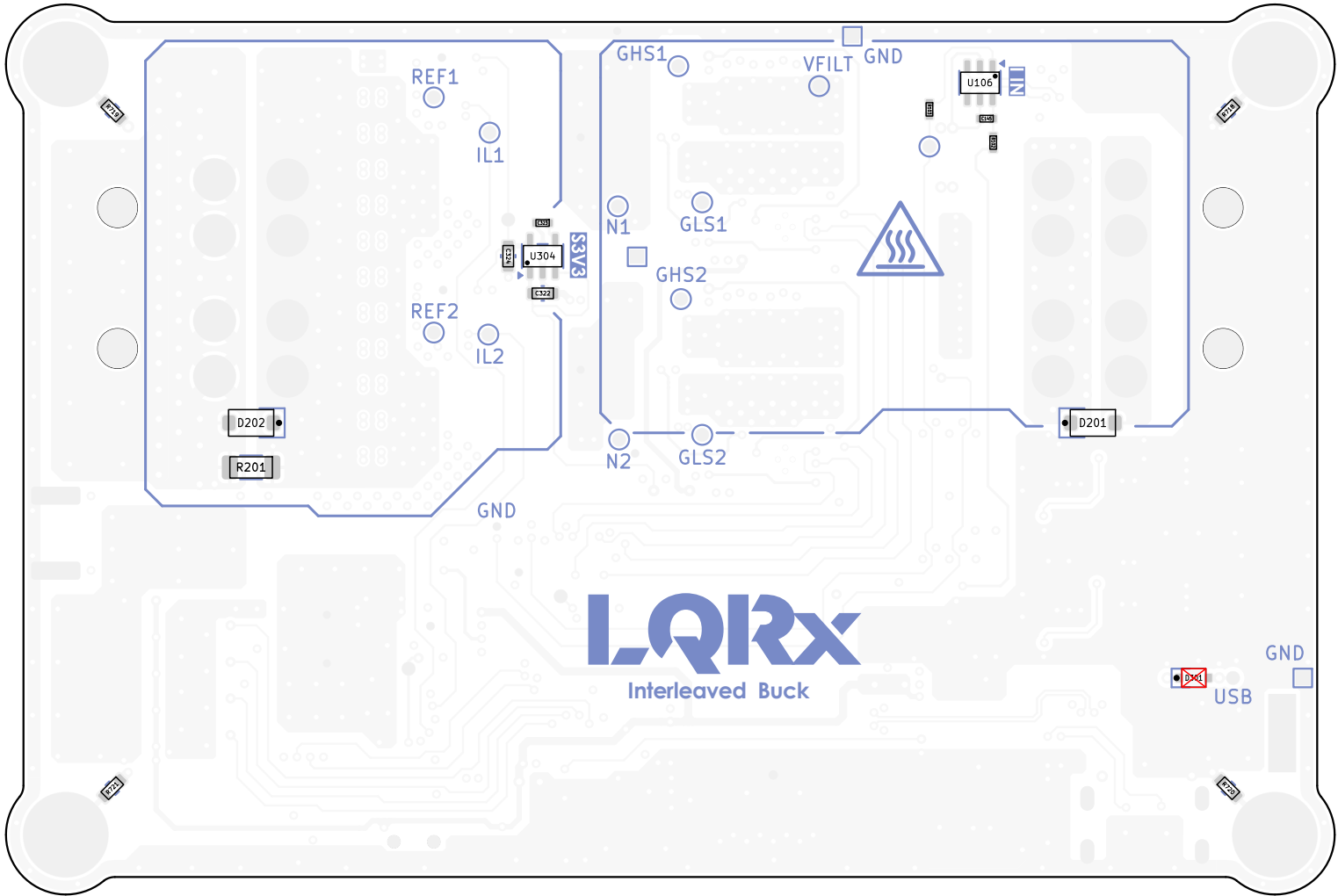
05-2026 V1.1

0 1 2 3 4 5

Project: 2-Phase Interleaved Sync-Buck Converter		LQRx	
Title: Fabrication Documents	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:
	Reviewed by:	Date:	Rev: 1.1 Date: 28/06/2026
Sheet: Assembly Top Side		Size: A4	Sheet: 2 of 5
KiCad E.D.A. 10.0.6		File: 2Phase_SynchBuck.kicad_pcb	

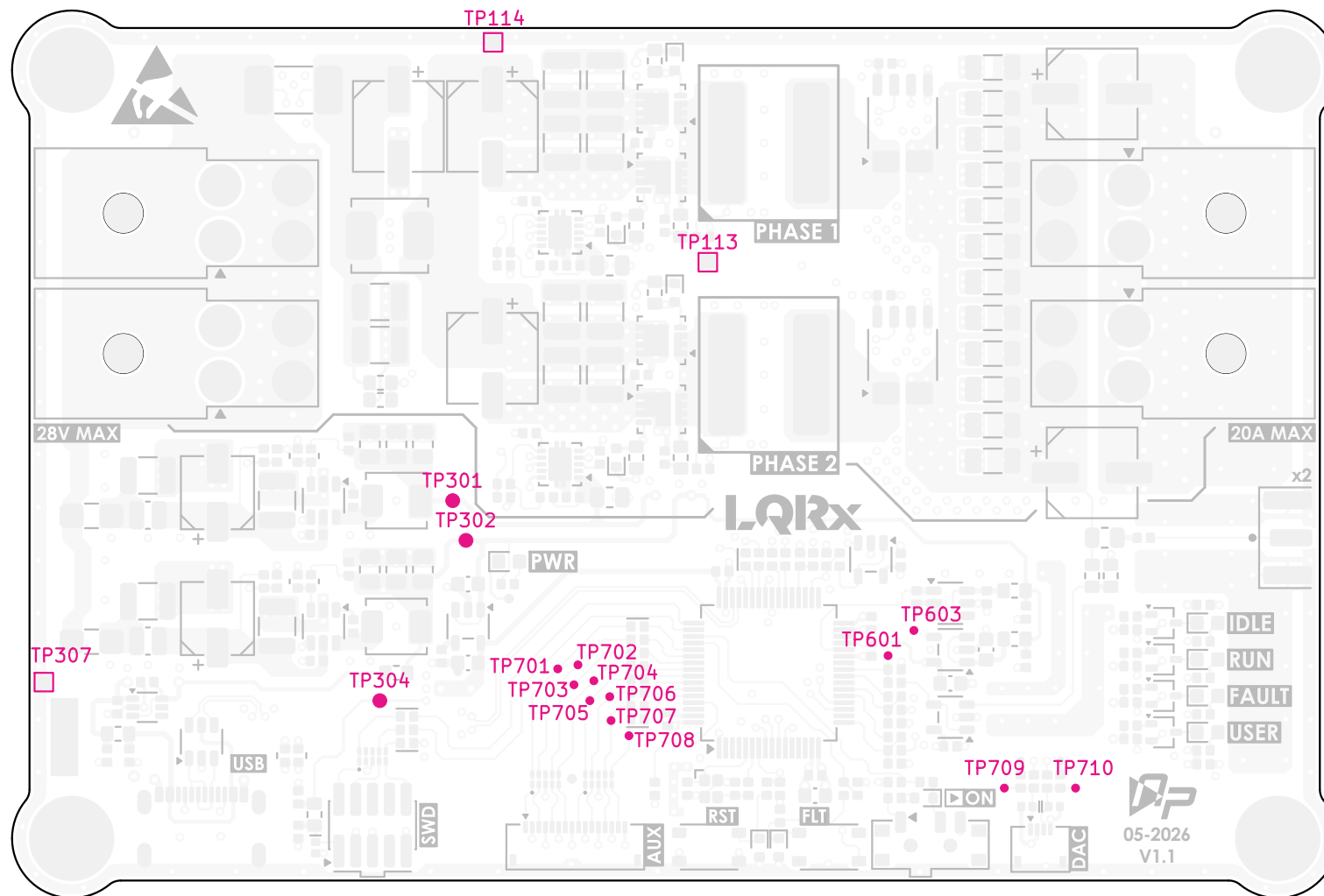
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Title: Fabrication Documents	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:
	Reviewed by:	Date:	Rev: 1.1 Date: 28/06/2026
	Sheet: Assembly Top Side		Size
	File: 2Phase_SynchBuck.kicad_pcb		Sheet: A4 2 of 5
KiCad E.D.A., 10.0.6			

Bottom Assembly (Scale 2:1)



Project: 2-Phase Interleaved Sync-Buck Converter		LQRx		
Title: Fabrication Documents	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:	
	Reviewed by:	Date:	Rev: 1.1	Date: 28/06/2026
	Sheet: Assembly Bottom Side		Size A4	Sheet: 3 of 5
KiCad E.D.A. 10.0.6		File: 2Phase_SynchBuck.kicad_pcb		

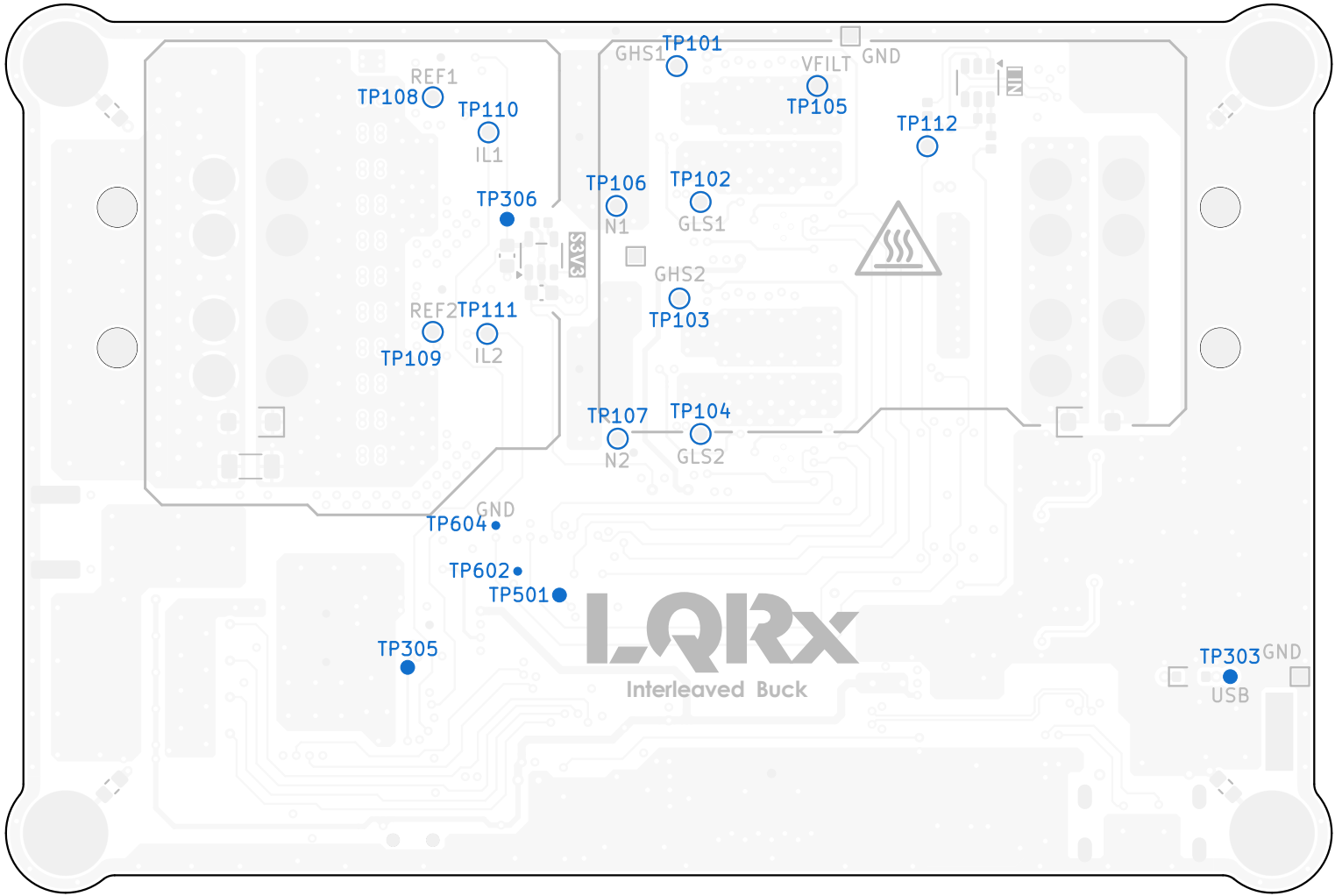
Top Testpoint (Scale 2:1)



Designator	Name
TP113	GND
TP114	GND
TP301	+9V
TP302	+5V
TP304	+3V3
TP307	GND
TP601	Analog Vout
TP603	GND
TP701	PC11
TP702	PC12
TP703	PD2
TP704	PB3
TP705	PB4
TP706	PB5
TP707	TX
TP708	RX
TP709	DAC1
TP710	DAC2

Project: 2-Phase Interleaved Sync-Buck Converter		LQRx		
Title: Fabrication Documents	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:	
	Reviewed by:	Date:	Rev: 1.1	Date: 28/06/2026
	Sheet: Testpoint Top Side		Size A4	Sheet: 4 of 5
KiCad E.D.A. 10.0.6		File: 2Phase_SynchBuck.kicad_pcb		

Bottom Testpoint (Scale 2:1)



Designator	Name
TP101	Gate HS 1
TP102	Gate LS 1
TP103	Gate HS 2
TP104	Gate LS 2
TP105	Vin Filter
TP106	Switch-node 1
TP107	Switch-node 2
TP108	VREF1
TP109	VREF2
TP110	IL1
TP111	IL2
TP112	I _{IN}
TP303	+USB
TP305	+A3V3
TP306	+S3V3
TP501	HTIM_E1
TP602	Analog VIN
TP604	GND

Project: 2-Phase Interleaved Sync-Buck Converter		LQRx			
Title: Fabrication Documents	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:		
	Reviewed by:	Date:	Rev: 1.1	Date: 28/06/2026	
	Sheet: Testpoint Bottom Side		Size A4	Sheet: 5 of 5	
KiCad E.D.A. 10.0.6	File: 2Phase_SynchBuck.kicad_pcb				